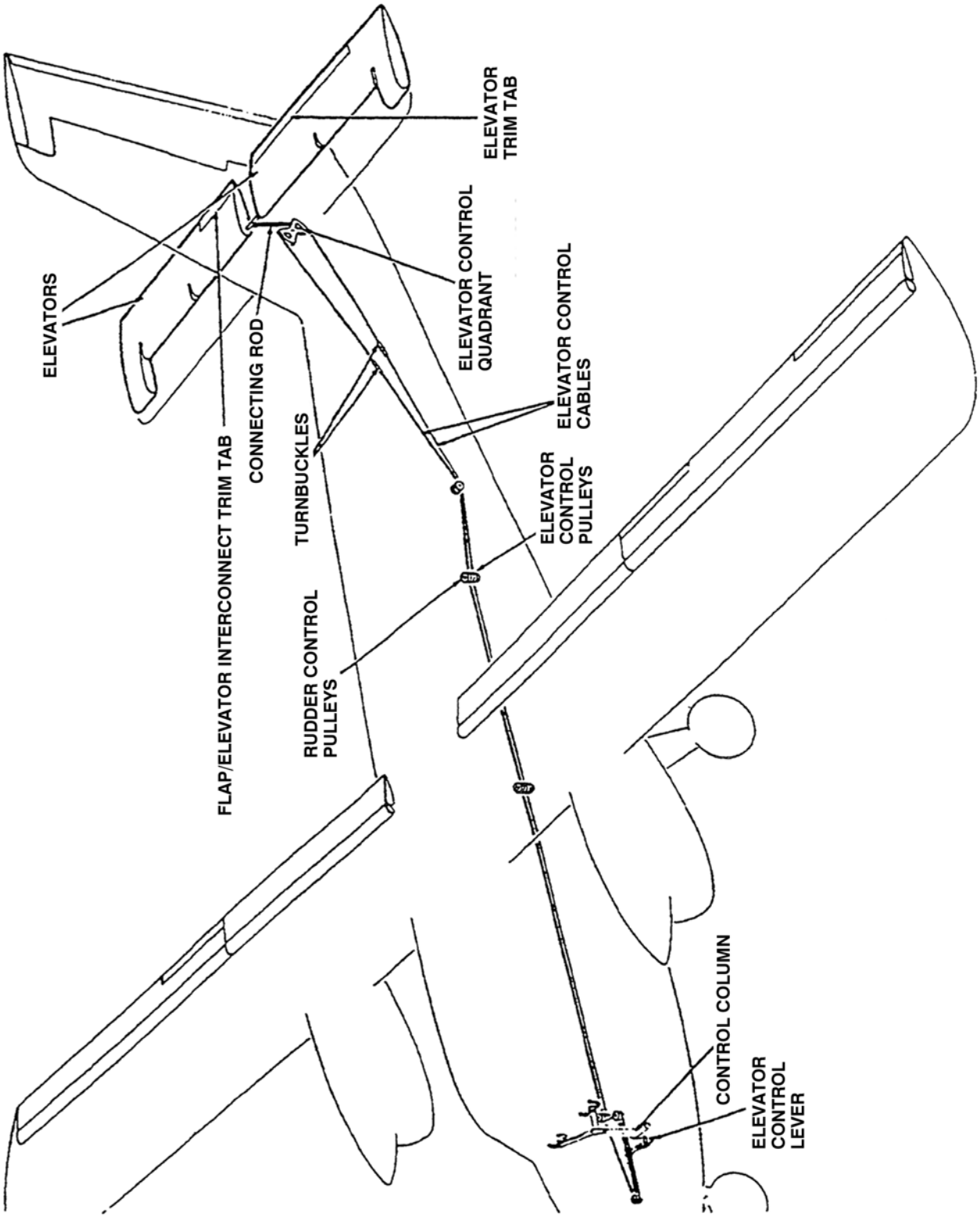
Elevator Control System – General Data

1. General

Elevator deflection is achieved through fore and aft movement of the control column. A connecting rod (located below the flight compartment floor) joins the control column to the elevator control lever which transmits movement to the elevator control cables. Movement of the elevator control lever is limited by a cable with swaged stops which passes through the outboard end of the control lever; the cable is secured at each end to elevator control pulley attachment bolts. On aircraft with Mod 6/1747 incorporated, end straps are installed on the stop cables to provide for cable tension adjustment. From the control lever, movement is transmitted through cables, routed through pulleys and fairleads on the right side of the fuselage under the cabin floor, to an elevator quadrant in the rear fuselage. An adjustable connecting rod transmits movement from the quadrant to the elevator torque tube and elevators. On aircraft with Mod 6/1739 incorporated, drainage holes are provided in the connecting rod. Two hinge attachments are mounted on the elevator front spar and a third is located on the flange at the end of the elevator torque tube. When the elevators are installed, the torque tube flanges, when aligned and bolted together, provide the maximum up and down travel stops. The left elevator carries an elevator trim tab and the right elevator carries a flap/elevator interconnect tab. Refer to Figure 1.

Note For general cable rigging instructions, refer to 27-00-00, Flight Controls – General Data.



Elevator Control System
Figure 1

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2. Adjustment/Test

Refer to Figure 2 for details.

A. Rig Elevator Control System

1) Lock control column in neutral by installing gust locks. On aircraft with Mod 6/1676 incorporated, obtain neutral by installing elevator rigging tool C6GT1047–1 and accompanying rigging pins C6GT1048–1 between control column and plate on instrument panel frame.

2) On aircraft with Mod 6/1747 incorporated, check stop cable for minimum slackness. Adjust tension of stop cable by using alternate mounting holes in straps to achieve minimum slackness, maintaining ease of insertion of stop cable mounting bolts. End strap hole pitch may be adjusted between +0.100 and –0.150 inch in increments of 0.050 inches.

Note Inspect elevator quadrant mounting bracket for cracks, and the quadrant for distortion and loose attachment bolts (refer to Inspection Requirements Manual PSM 1-6-7, Basic and Special Inspections).

3) Check that top surface of right elevator horn and lower edge of rigging disc at horizontal stabilizer tip are aligned within ± 0.030 inches (neutral position).

Note If Para 2.A.3) is satisfactory, ignore Para 2.A.4), Para 2.A.5) and Para 2.A.6).

4) Insert rigging pin SD10542 (0.25 inch dia.) in elevator control quadrant in rear fuselage.

a) If rigging pin enters easily, adjust connecting rod at elevator as necessary to comply to Para 2.A.3).

b) If rigging pin does not enter quadrant, adjust turnbuckles in rear fuselage until quadrant setting is obtained; tension cables to values given in 27-00-00, Flight Controls – General Data.

5) Check rigging disc at right elevator horn for alignment, as in Para 2.A.3), and make final adjustment on connecting rod as necessary. Safety connecting rod. Lock turnbuckles using locking clips as necessary.

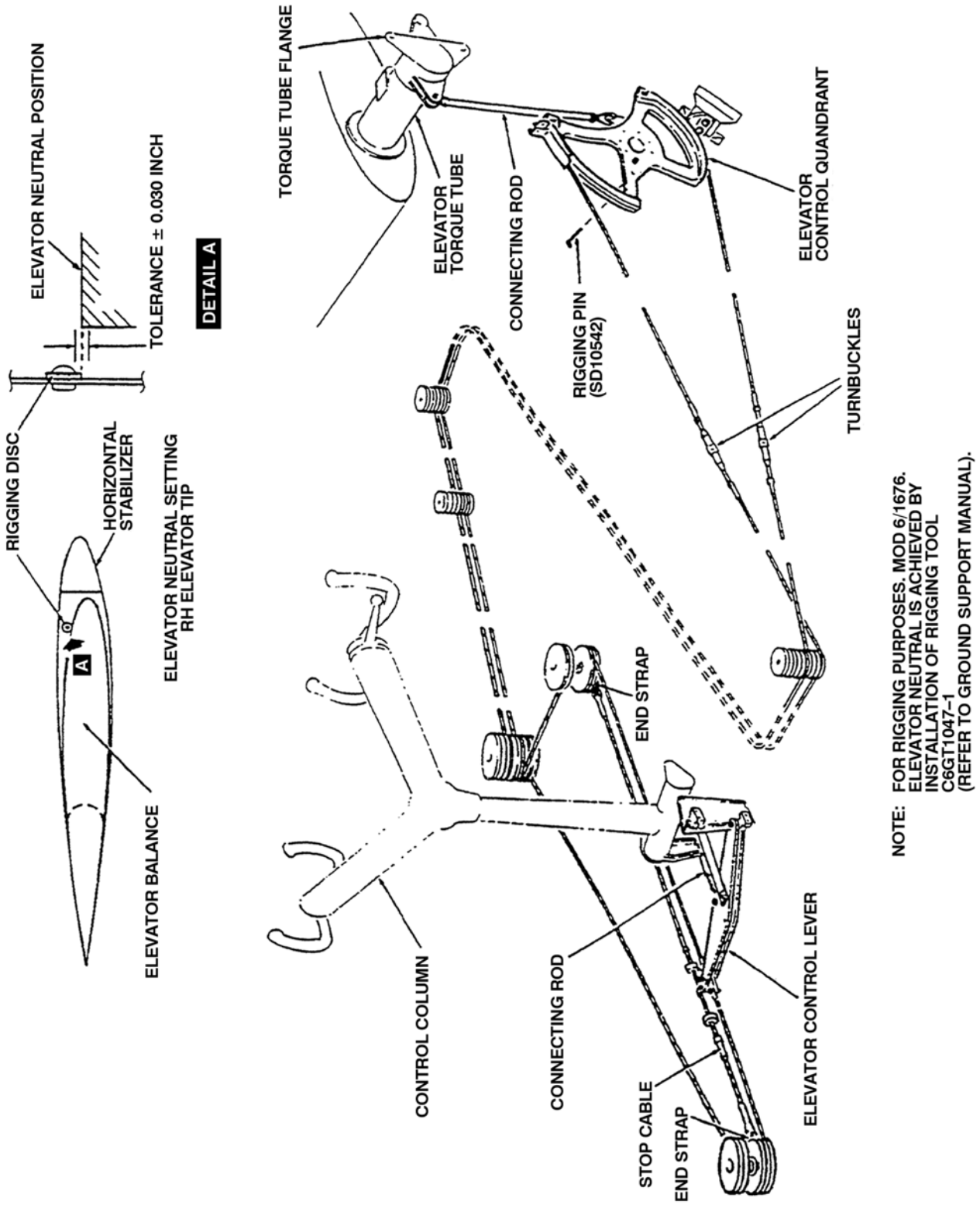
6) Remove rigging pins.

7) Remove rigging tool (Mod 6/1676).

8) Check elevator control surface clearances (refer to 27-00-00, Flight Controls – General Data).

9) Operate elevator controls and check for full and free range of movement.

10) Check rigging of elevator trim tab and flap/elevator interconnect trim tab systems (refer to 27-33-00, Elevator Trim Tab System – General Data and 27-36-00, Flap/Elevator Interconnect Trim System – General Data).



Elevator System Rigging
Figure 2

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3. Inspection/Check

A. Job Set-up

1) Either pull and clip breakers required to isolate affected circuits or placard and select aircraft electrical power to OFF.

2) Obey all relevant WARNINGS and CAUTIONS detailed in Maintenance Manual PSM 1-64-2, Chapters 12 and 24.

3) Placard flight controls in flight compartment.

B. Flight Compartment

1) Open access doors 15 to gain access to elevator cables (refer to Table 1 — Elevator Cable Assy – Flight Compartment and Passenger Cabin and Figure 2).

2) Do a visual inspection of control cables (non-jacketed) visible at this section of fuselage (refer to Table 1 — Elevator Cable Assy – Flight Compartment and Passenger Cabin and Figure 2). Observe carefully areas where cables pass over pulleys, quadrants, drums, or through fairleads.

3) Replace cables not acceptable to criteria in Inspection of Control Cables (refer to 20-60-01, Inspection, Cleaning and Lubrication of 7x19 and 7x7 Control Cables – General Data). Make sure the new cable part numbers are identical to the discarded cable part numbers (refer to Table 1 — Elevator Cable Assy – Flight Compartment and Passenger Cabin).

4) Lubricate new or existing non-jacketed Carbon Steel cables (refer to 20-60-01, Inspection, Cleaning and Lubrication of 7x19 and 7x7 Control Cables – General Data).

Note Do not lubricate stainless steel cables.

5) Replace cable as required.

6) Install and rig replaced cable assemblies (refer to 27-00-00, Flight Controls – General Data).

7) Close access doors 15.

Table 1 Elevator Cable Assy – Flight Compartment and Passenger Cabin

	CARBON STEEL	STAINLESS STEEL
Front of elevator lever to station 426.75	C6CF1146–1	E.O. 69053–1
Front of elevator lever to station 426.75	C6CF1147–1	E.O. 69053–2
Elevator lever stop cable	C6CF1100–11	E.O. 69053–5

C. Passenger Cabin

1) Remove panel 43 (refer to 12-00-00, Servicing – General Data, Figure 1) to gain access to elevator cables and pulleys (refer to Table 1 — Elevator Cable Assy – Flight Compartment and Passenger Cabin and Figure 2).

2) Release, attach string and pull back elevator cables to allow visual inspection (refer to Table 1 — Elevator Cable Assy – Flight Compartment and Passenger Cabin, Figure 2 and Inspection of Control Cables PSM 1-64-2, Section 27-00-00).

3) Observe carefully areas where cables pass over pulleys, quadrants, drums, or through fairleads.

4) Replace cables not acceptable to criteria in Inspection of Control Cables (refer to 27-00-00, Flight Controls – General Data). Make sure the new cable part numbers are identical to the discarded cable part numbers per Table 1 — Elevator Cable Assy – Flight Compartment and Passenger Cabin.

5) Lubricated replaced or existing non-jacketed Carbon Steel cables (refer to 20-60-01, Inspection, Cleaning and Lubrication of 7x19 and 7x7 Control Cables – General Data).

Note Do not lubricate stainless steel cables.

6) Install and rig cable assemblies (refer to 27-00-00, Flight Controls – General Data).

7) Install panel 43.

D. Rear Fuselage and Tail Area

CAUTION **WHEN INSTALLING ELEVATOR AND RUDDER PRIMARY FLIGHT CONTROL CABLES, CARGO AREA FLOOR BOARDS (BETWEEN STA 332.00 TO 376.00 AND AFT CARGO AREA FLOOR BOARDS (AFT OF STA 376.00, MUST BE REMOVED TO ENSURE PROPER INSTALLATION. THIS WILL REDUCE THE POSSIBILITY OF MISS-RIGGING AND CABLE CROSSING, WHICH COULD LEAD TO ACCELERATED CABLE WEAR TO THE ELEVATOR AND RUDDER PRIMARY FLIGHT CONTROL SYSTEMS.**

1) Remove floor panels to gain access to elevator cable assemblies in baggage compartment rear fuselage.

2) Remove panels 11 and 12 (refer to Figure 2) to gain access to elevator cable assemblies in the tail area.

3) Do a visual inspection of control cables (non-jacketed) visible at this section of fuselage (refer to Table 2 — Elevator Cable Assy – Rear Fuselage and Tail Area, Figure 2 and 27-00-00, Flight Controls – General Data, Para 3.A). Observe carefully areas where cables pass over pulleys, quadrants, drums, or through fairleads and cable ends. Place special emphasis on elevator and rudder cable inspection near cable cluster at station 376.0 (refer to Figure 2).

4) Replace cables not acceptable to criteria in Inspection of Control Cables (refer to 27-00-00, Flight Controls – General Data). Make sure the new cable part numbers are identical to the discarded cable part numbers per Table 2 — Elevator Cable Assy – Rear Fuselage and Tail Area.

5) Lubricate new or existing non-jacketed Carbon Steel cables only (refer to 20-60-01, Inspection, Cleaning and Lubrication of 7x19 and 7x7 Control Cables – General Data).

Note Do not lubricate stainless steel cables.

6) Install and rig replaced cable assemblies (refer to 27-00-00, Flight Controls – General Data).

7) Install baggage compartment floor panels and panels 11 and 12.

Table 2 Elevator Cable Assy – Rear Fuselage and Tail Area

	CARBON STEEL	STAINLESS STEEL
Upper cable, station 426.75 to elevator quadrant	C6CF1148–1	E.O. 69053–3
Lower cable, station 426.75 to elevator quadrant	C6CF1149–1	E.O. 69053–4

E. Close-out:

1) Remove placards on the flight controls in flight compartment.

2) Obey all relevant WARNINGS and CAUTIONS detailed in Maintenance Manual PSM 1-64-2, Chapters 12 and 24.

3) Either remove clip and push breakers that were required to isolate affected circuits or remove placards and select aircraft electrical power to ON.

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